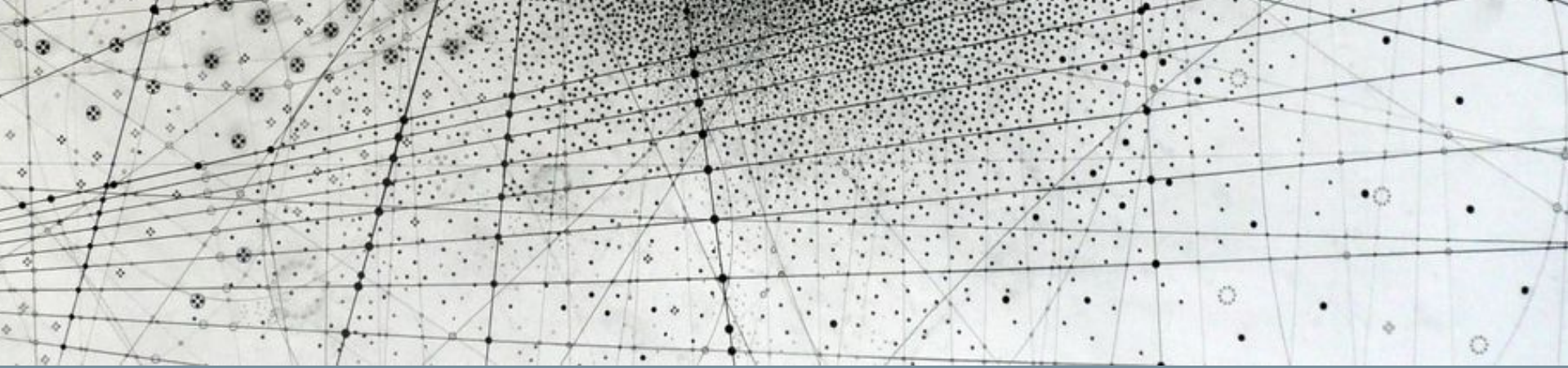




Research Methods in Machine Learning & Physics

Announcements

Project #2 will be posted this week

Problem Set #2 solutions posted – come chat at office hours tomorrow if you'd like to discuss them!

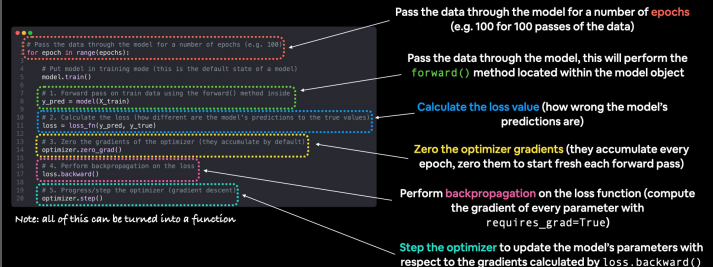
- I will adjust scores for e.g. 60% vs 0.6
- I'll also give more generous credit for the following problem:

Problem 4: Order of operations during training

Put the following operations in order and add them to the training loop below.

- a) `loss.backward()`
- b) `optimizer.step()`
- c) `loss = criterion(out, data.y.float())`
- d) `out = model(data.x, data.edge_index, data.batch).squeeze()`
- e) `optimizer.zero_grad()`

PyTorch training loop



https://docs.pytorch.org/tutorials/beginner/basics/optimization_tutorial.html

https://github.com/mrdbourke/pytorch-deep-learning/blob/main/01_pytorch_workflow.ipynb

For the training loop, we'll build the following steps:

Number	Step name	What does it do?	Code example
1	Forward pass	The model goes through all of the training data once, performing its <code>forward()</code> function calculations.	<code>model(x_train)</code>
2	Calculate the loss	The model's outputs (predictions) are compared to the ground truth and evaluated to see how wrong they are.	<code>loss = loss_fn(y_pred, y_train)</code>
3	Zero gradients	The optimizers gradients are set to zero (they are accumulated by default) so they can be recalculated for the specific training step.	<code>optimizer.zero_grad()</code>
4	Perform backpropagation on the loss	Computes the gradient of the loss with respect for every model parameter to be updated (each parameter with <code>requires_grad=True</code>). This is known as backpropagation , hence "backwards".	<code>loss.backward()</code>
5	Update the optimizer (gradient descent)	Update the parameters with <code>requires_grad=True</code> with respect to the loss gradients in order to improve them.	<code>optimizer.step()</code>

```
def train_loop(dataloader, model, loss_fn, optimizer):
    size = len(dataloader.dataset)
    # Set the model to training mode - important for batch normalization and dropout layers
    # Unnecessary in this situation but added for best practices
    model.train()
    for batch, (X, y) in enumerate(dataloader):
        # Compute prediction and loss
        pred = model(X)
        loss = loss_fn(pred, y)

        # Backpropagation
        loss.backward()
        optimizer.step()
        optimizer.zero_grad()

    if batch % 100 == 0:
        loss, current = loss.item(), batch * batch_size + len(X)
        print(f"loss: {loss:>7f}   [{current:>5d}/{size:>5d}"]
```



Daniel Bourke

@mrdbourke

Let's sing the @PyTorch optimization loop song 🎵

It's train time!
do the forward pass,
calculate the loss,
optimizer zero grad,
losssss backwards!

Optimizer step step step

Let's test now!
with torch no grad:
do the forward pass,
calculate the loss,
watch it go down down down!

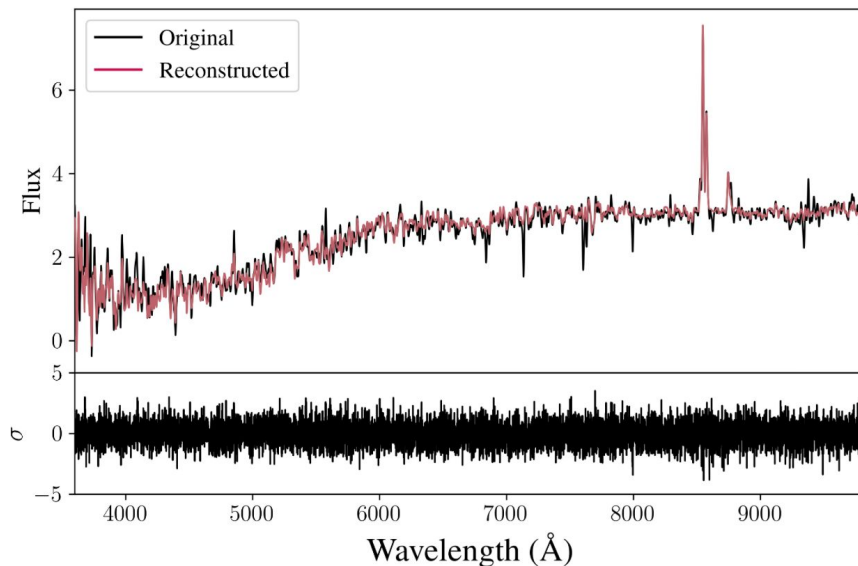
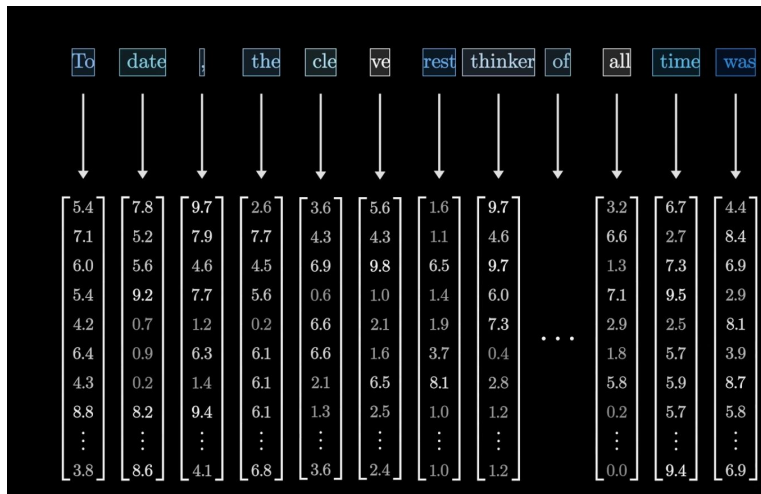


Quick recap

Tokenization

Tokenizers convert your data into fixed-sized “tokens” to feed to your Transformer.

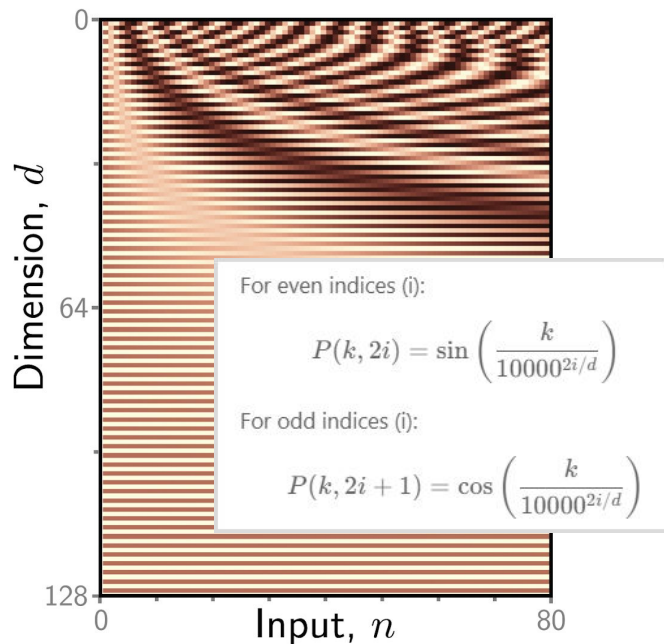
You can use pre-trained tokenizers for e.g. text/images, but might have to train your own for specialized data.



Positional encodings

Absolute

e.g. sinusoidal:



Relative

- RoPE = Rotary Position Embedding

ROFORMER: ENHANCED TRANSFORMER WITH ROTARY POSITION EMBEDDING

Jianlin Su
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Yunfeng Liu
Zhuiyi Technology Co., Ltd.
Shenzhen
glenliu@wezhuiyi.com

November 9, 2023

- ALiBi = Attention with Linear Biases

TRAIN SHORT, TEST LONG: ATTENTION WITH LINEAR BIASES ENABLES INPUT LENGTH EXTRAPOLATION

Ofir Press^{1,2} Noah A. Smith^{1,3} Mike Lewis²

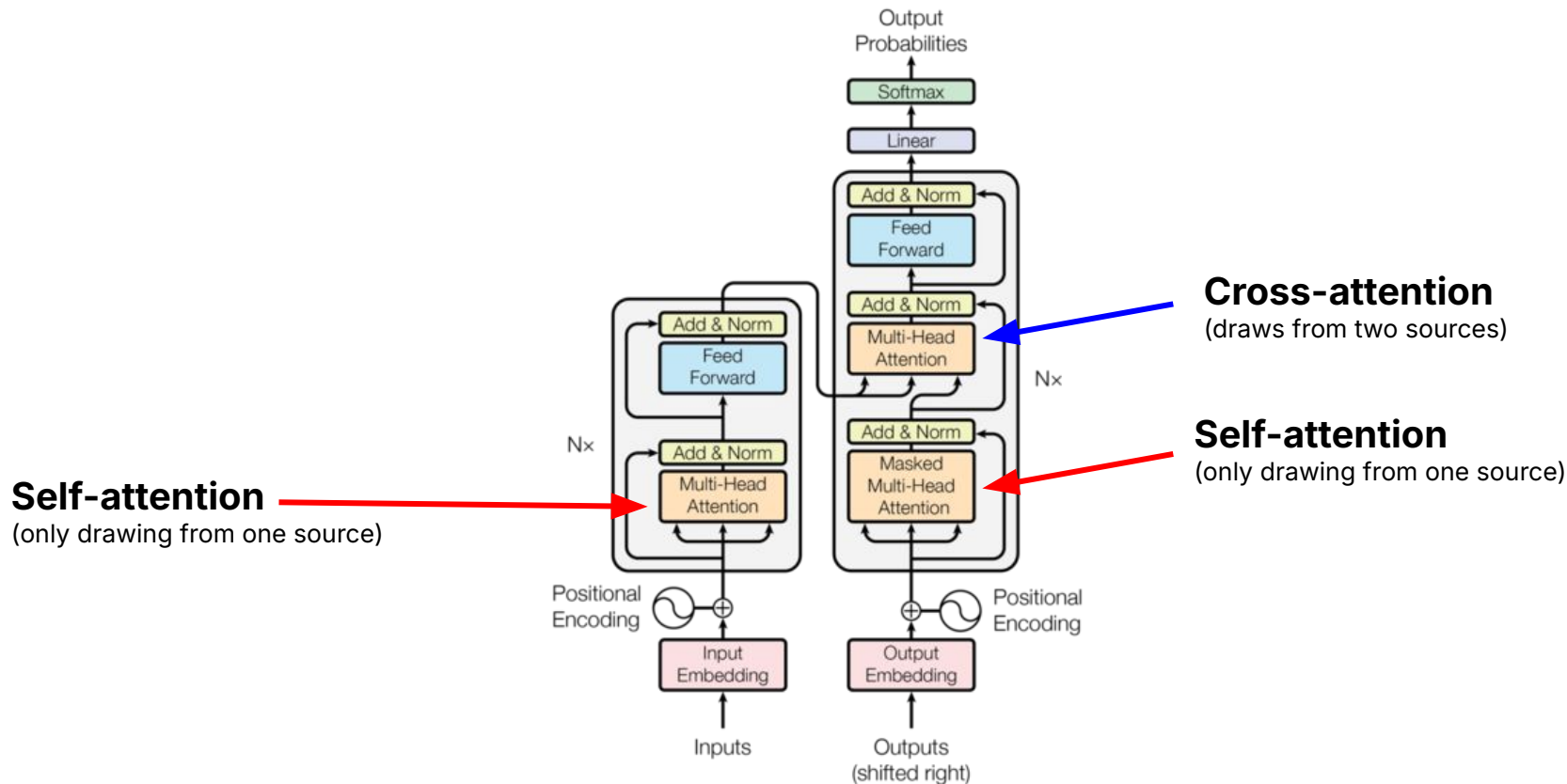
¹Paul G. Allen School of Computer Science & Engineering, University of Washington

²Facebook AI Research

³Allen Institute for AI

ofirp@cs.washington.edu

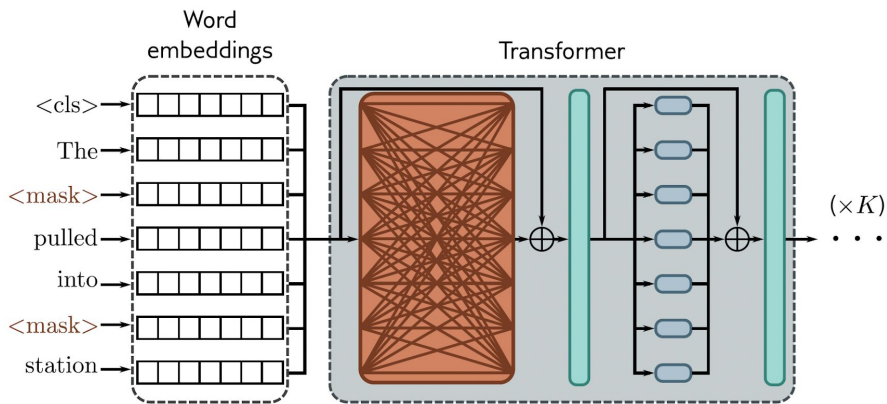
Self-attention & cross-attention



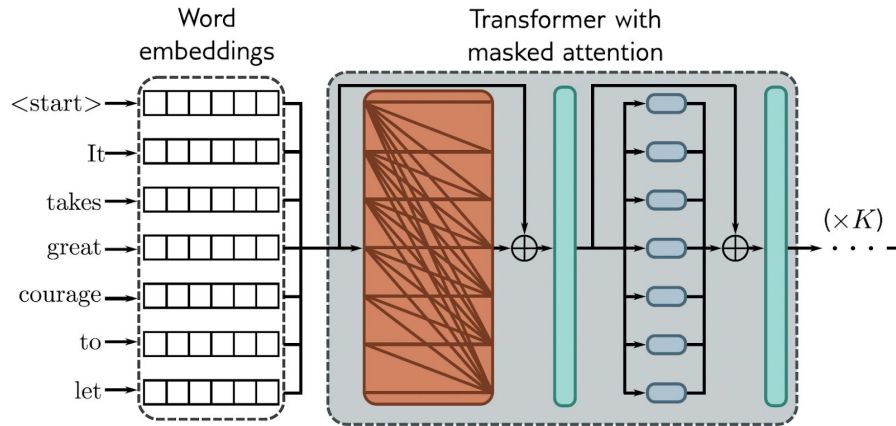
LLMs

Transformers → Large Language Models (LLMs)

Encoder

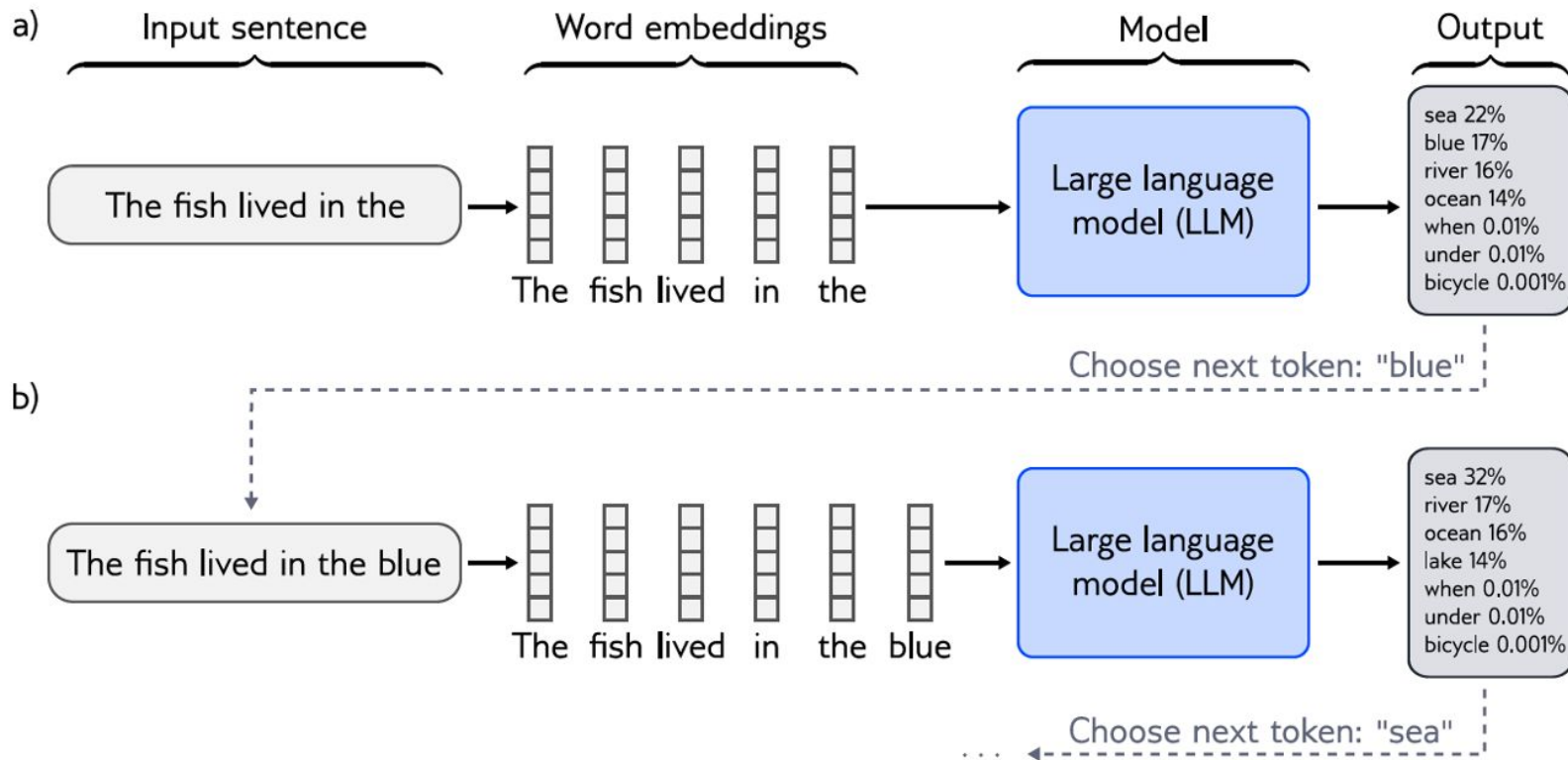


Decoder



Encoder-Decoder (translation) – put these two together

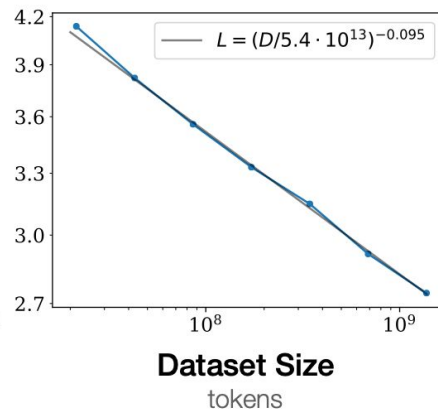
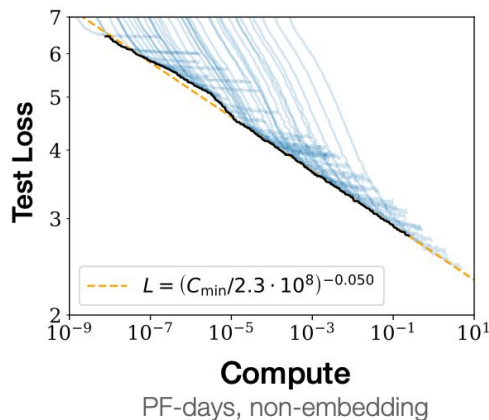
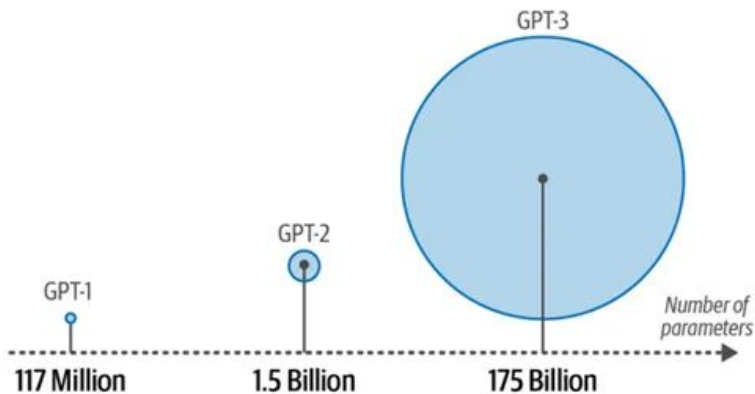
Decoder-only = predict the next token



Transformers → Large Language Models (LLMs)

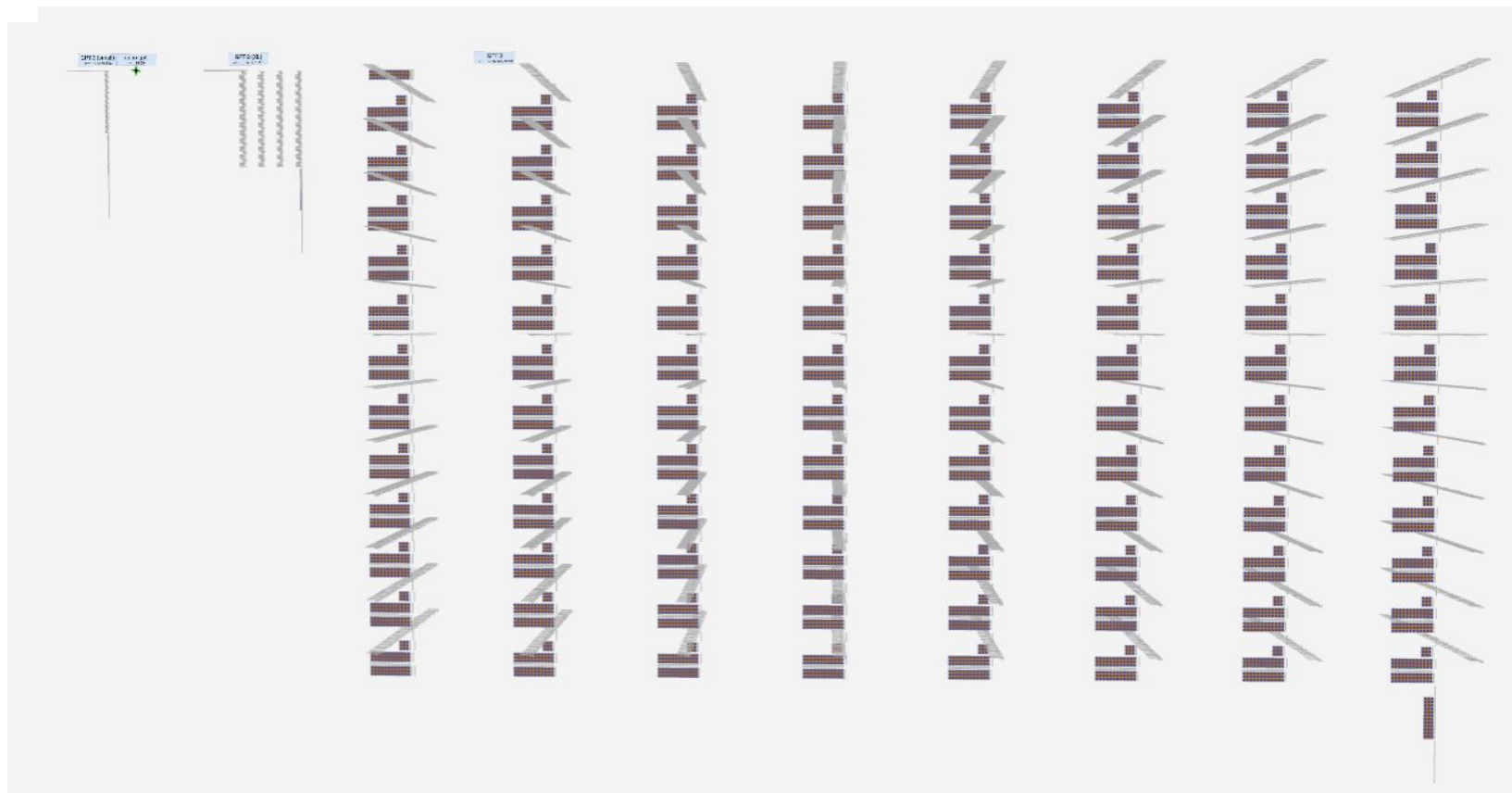
The notion of Large Language Models combines the decoder-only Transformer architecture with a huge amount of data & compute.

Scaling laws seemingly indicate that model performance will increase forever with more compute:



(Training dataset size: 300 billion tokens!)

Transformers → Large Language Models (LLMs)

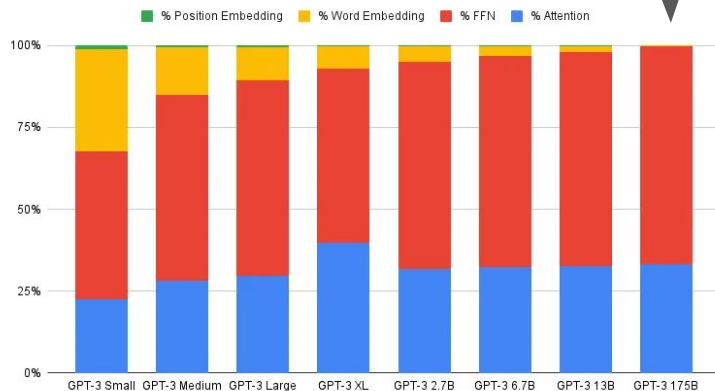


GPT-3 (2020)

Note that the parameter count of GPT-3 is mostly dedicated toward the MLP layers, not the attention layers:

Model Name	n_{params}	n_{layers}	d_{model}	n_{heads}	d_{head}
GPT-3 Small	125M	12	768	12	64
GPT-3 Medium	350M	24	1024	16	64
GPT-3 Large	760M	24	1536	16	96
GPT-3 XL	1.3B	24	2048	24	128
GPT-3 2.7B	2.7B	32	2560	32	80
GPT-3 6.7B	6.7B	32	4096	32	128
GPT-3 13B	13.0B	40	5140	40	128
GPT-3 175B or "GPT-3"	175.0B	96	12288	96	128

GPT-3 Parameter Usage



About 1/3 of What
Attention is ~~All~~ You Need

LLMs for physics

LLMs for physics

Why would we care about LLMs in the context of physics analyses? We're not analyzing language data!

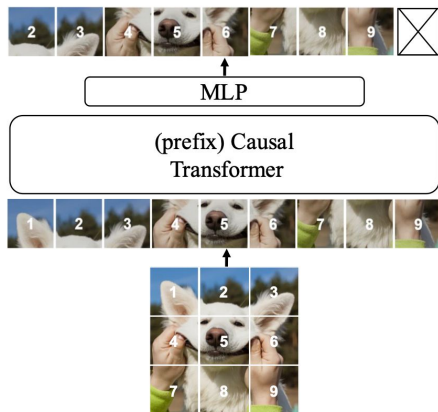
Reason 1: autoregressive next-token prediction is a useful framework beyond language

Scalable Pre-training of Large Autoregressive Image Models

Alaaeldin El-Nouby Michal Klein Shuangfei Zhai Miguel Angel Bautista
Alexander Toshev Vaishaal Shankar Joshua M Susskind Armand Joulin*

Apple

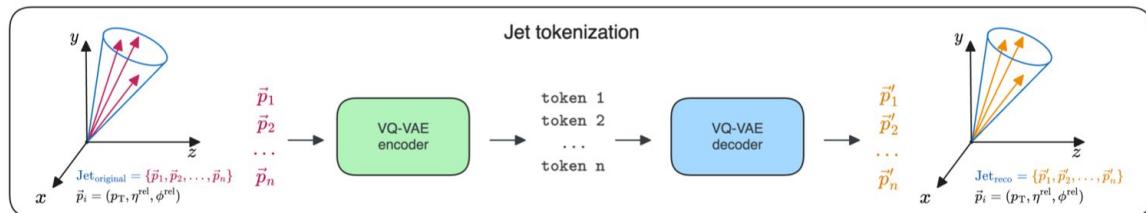
<https://github.com/apple/ml-aim>



OmniJet- α : The first cross-task foundation model for particle physics

Joschka Birk,^{1,*} Anna Hallin,^{1,†} and Gregor Kasieczka¹

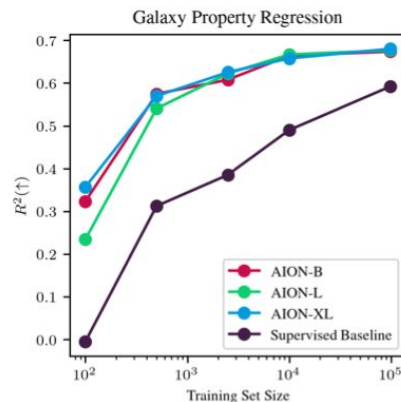
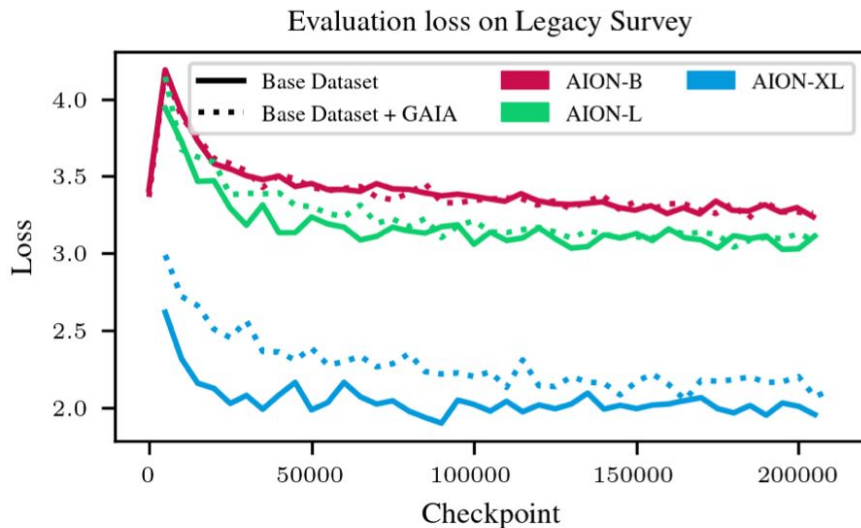
¹*Institute for Experimental Physics, Universität Hamburg
Luruper Chaussee 149, 22761 Hamburg, Germany*



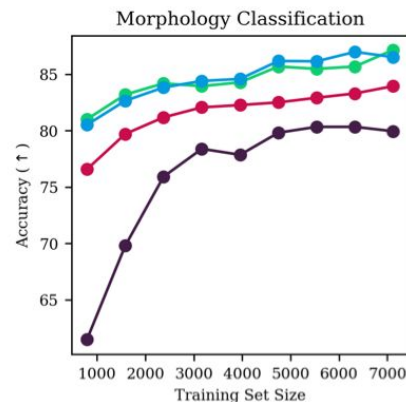
LLMs for physics

Why would we care about LLMs in the context of physics analyses? We're not analyzing language data!

Reason 2: scaling laws are critical in scientific ML, too



(a): Galaxy Properties



(b): Morphology Classification

LLMs for physics

Why would we care about LLMs in the context of physics analyses? We're not analyzing language data!

Reason 3: actually, maybe language *is* relevant for physics analyses

PAPERCLIP: Associating Astronomical Observations and Natural Language with Multi-Modal Models

Siddharth Mishra-Sharma

The NSF AI Institute for Artificial Intelligence and Fundamental Interactions
Center for Theoretical Physics, Massachusetts Institute of Technology, Cambridge, MA 02139, USA
Department of Physics, Harvard University, Cambridge, MA 02138, USA

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Yiding Song

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Jesse Thaler

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Center for Theoretical Physics, Massachusetts Institute of Technology, Cambridge, MA 02139, USA

jthaler@mit.edu

CLIP fine-tuning

With Hubble *observation-proposal abstract pairs*

Hubble proposal abstracts

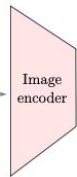
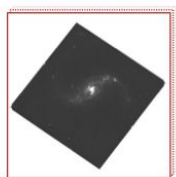
Category: Galaxies. We propose WFC3/UVIS F336W, F438W, and F814W observations for 8 Luminous Infrared Galaxies (LIRGs) in the Great Observatories All-Sky LIRG Survey (GOALS) scheduled for JWST Cycle 1 (GO1) observations. With a proprietary period of 8 days for 50% of the GO1 LIRGs, observations taken now will provide the concurrent WFC3/UVIS imaging necessary to reliably age-date the star ...

Mixtral + Outlines

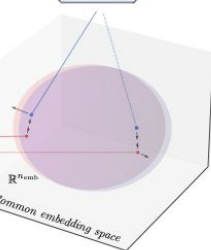
Optional summarization with constrained LLM generation

Luminous Infrared Galaxies, star clusters, nuclear regions, extranuclear regions, hydrogen recombination lines; measure fraction of star formation in clusters, determine nuclear and extranuclear cluster destruction rates, ...

Hubble observations



Text encoder

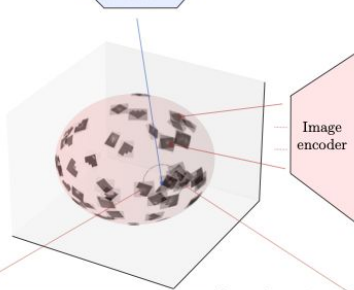


Downstream task: observation retrieval

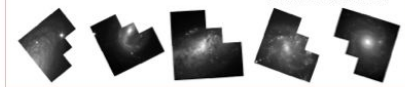
Given natural language text query

Query: "barred spiral galaxy"

Text encoder



Closest observations



LLMs for physics

Why would we care about LLMs in the context of physics analyses? We're not analyzing language data!

Reason 3: actually, maybe language *is* relevant for physics analyses

Large Language Models — the Future of Fundamental Physics?

Caroline Heneka¹, Florian Nieser^{2,3}, Ayodele Ore¹, Tilman Plehn^{1,3}, and Daniel Schiller¹

¹ Institut für Theoretische Physik, Universität Heidelberg, Germany

² Heidelberg Center for Digital Humanities (HCDH), Universität Heidelberg, Germany

³ Interdisciplinary Center for Scientific Computing (IWR), Universität Heidelberg, Germany

June 18, 2025

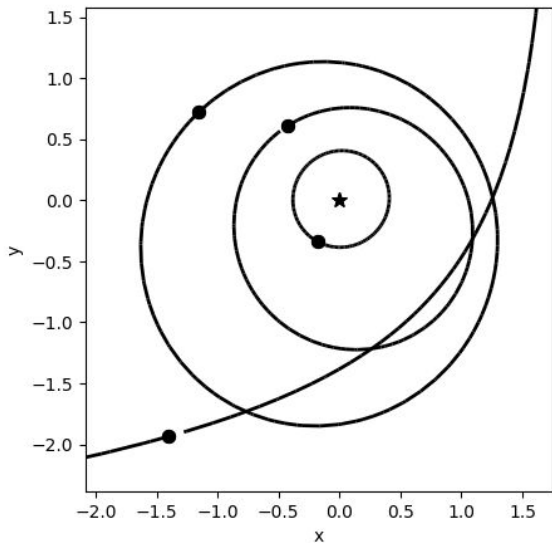
Abstract

For many fundamental physics applications, transformers, as the state of the art in learning complex correlations, benefit from pretraining on quasi-out-of-domain data. The obvious question is whether we can exploit Large Language Models, requiring proper out-of-domain transfer learning. We show how the Qwen2.5 LLM can be used to analyze and generate SKA data, specifically 3D maps of the cosmological large-scale structure for a large part of the observable Universe. We combine the LLM with connector networks and show, for cosmological parameter regression and lightcone generation, that this Lightcone LLM (L3M) with Qwen2.5 weights outperforms standard initialization and compares favorably with dedicated networks of matching size.

LLMs for physics

Why would we care about LLMs in the context of physics analyses? We're not analyzing language data!

Reason 4: using LLMs for multimodal embeddings & data analysis



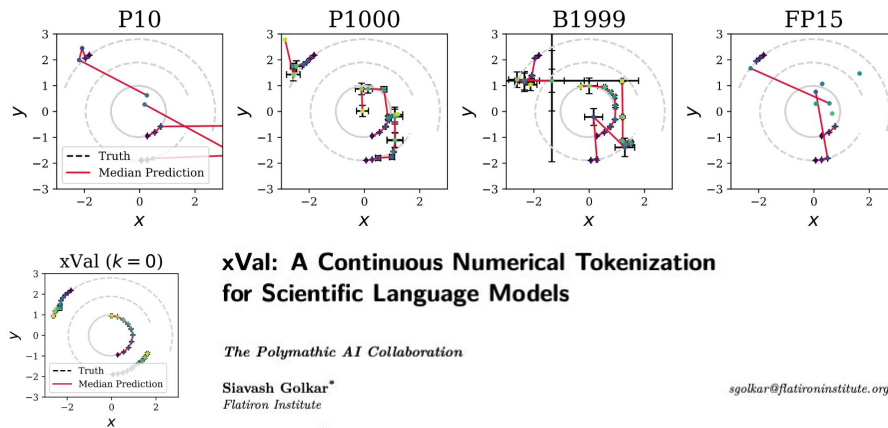
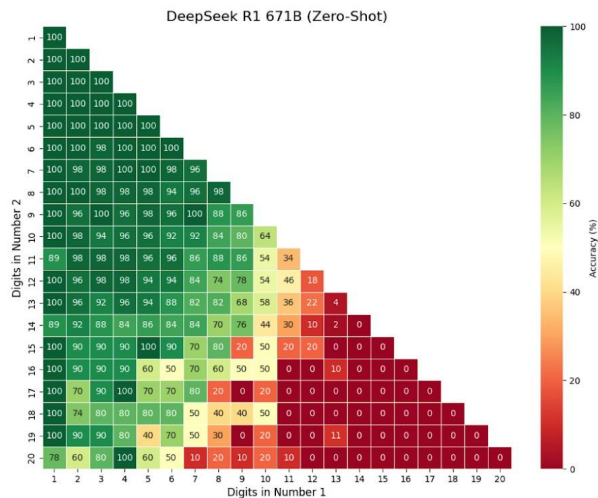
```
{ 'description': {  
  'planet0': {  
    'm': 4.842, 'a': 2.679, 'e': 0.923  
  },  
  'planet1': {  
    'm': 4.813, 'a': 1.0, 'e': 0.191  
  },  
  'stepsize': 0.8,  
  'data': [  
    [[2.525, -0.397], [0.963, -0.232]],  
    [[2.557, 0.112], [0.830, 0.542]],  
    [[2.483, 0.618], [0.1832, 0.981]],  
    [[2.318, 1.100], [-0.577, 0.823]],  
    [[2.066, 1.541], [-0.994, 0.173]]  
  ]  
}
```

LLMs for physics

Why would we care about LLMs in the context of physics analyses? We're not analyzing language data!

Reason 4: using LLMs for multimodal embeddings & data analysis

LLMs are pretty bad at arithmetic out-of-the-box, but for physics analyses, we care more about continuous predictions. Specialized continuous embeddings can yield big improvements on physics data:



xVal: A Continuous Numerical Tokenization for Scientific Language Models

The Polymathic AI Collaboration

Siavash Golkar*
Flatiron Institute

Marcel Petee*
Lawrence Berkeley National Laboratory

sgolkar@flatironinstitute.org

mpetee@flatironinstitute.org

(GPT-4 fails at 4-digit multiplication over 95% of the time.)

LLMs for physics

Why would we care about LLMs in the context of physics analyses? We're not analyzing language data!

Reason 4: using LLMs for multimodal embeddings & data analysis

Direct data analysis with LLMs is better than you might expect, but still quite brittle and unreliable:

Large Language Models Are Zero-Shot Time Series Forecasters

Nate Gruver*
NYU

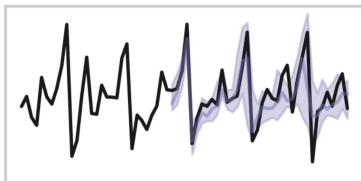
Marc Finzi*
CMU

Shikai Qiu*
NYU

Andrew Gordon Wilson
NYU

"1 5 1, 1 6 7, ... , 2 6 7"

"1 5 1, 1 6 7, ... , 2 6 7"

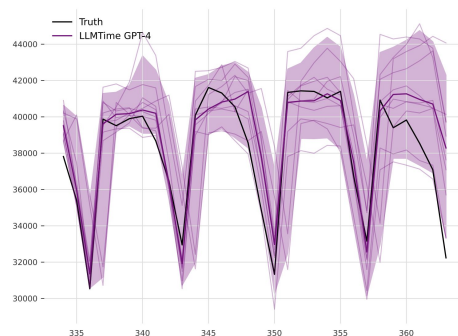


■ GPT-3 spaces

Time Series Forecasting with LLMs: Understanding and Enhancing Model Capabilities

Mingyu Jin^{1,*}, Hua Tang^{2,*}, Chong Zhang^{3,*}, Qinkai Yu³, Chengzhi Liu³,
Suiyuan Zhu⁵, Yongfeng Zhang¹, Mengnan Du⁴

¹ Rutgers University, ² Shanghai Jiao Tong University, ³ University of Liverpool
⁴ New Jersey Institute of Technology, ⁵ New York University



Just Because You Can, Doesn't Mean You Should:

LLMs for Data Fitting

Hejia Liu¹, Mochen Yang¹, Gediminas Adomavicius¹

¹ Carlson School of Management, University of Minnesota

Current Draft: 8/26/2025

learning task can drastically alter LLMs' predictions on the same data. [For example, simply changing variable names can sway the size of prediction error by as much as 82% in certain settings.](#) Such prediction sensitivity with respect to task-irrelevant variations manifests under both in-context learning and supervised fine-tuning, for both close-weight and open-weight general-purpose LLMs. Moreover, by

LLMs for physics

Why would we care about LLMs in the context of physics analyses? We're not analyzing language data!

Reason 5: LLMs for physics theory / proof-solving / interpretability

TP Bench – Theoretical Physics Benchmark for AI

TPBench is a curated dataset and evaluation suite designed to measure the reasoning capabilities of AI models in theoretical physics. Our test problems span multiple difficulty levels—from undergraduate to frontier research—and cover topics such as cosmology, high-energy theory, general relativity, and more. By providing a unified framework for problem-solving and auto-verifiable answers, TPBench aims to drive progress in AI-based research assistance for theoretical physics.

Theoretical Physics Benchmark (TPBench) - a Dataset and Study of AI Reasoning Capabilities in Theoretical Physics

Daniel J.H. Chung¹, Zhiqi Gao², Yuri Kvasiuk¹, Tianyi Li¹, Moritz Münchmeyer^{1,5}, Maja Rudolph³, Frederic Sala², and Sai Chaitanya Tadepalli⁴

¹Department of Physics, University of Wisconsin-Madison

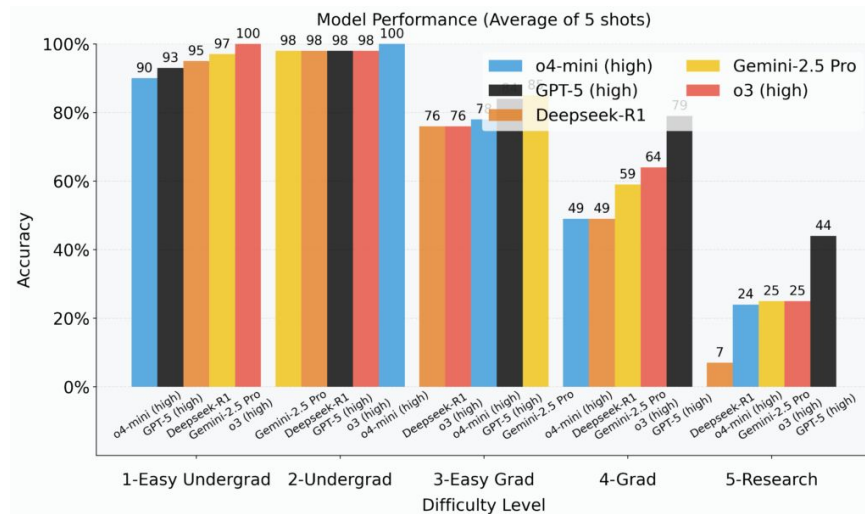
²Department of Computer Science, University of Wisconsin-Madison

³Data Science Institute (DSI), University of Wisconsin-Madison

⁴Department of Physics, Indiana University, Bloomington

⁵NSF-Simons AI Institute for the Sky (SkAI), Chicago

February 25, 2025



Additional resources

Andrej Karpathy's tutorials:

Let's build GPT: from scratch, in code, spelled out.
Deep Dive into LLMs like ChatGPT

LLM Visualizations: <https://bbycroft.net/llm>

In-class work

One person from each group will present their solution to the class:

Task 1: Tokenize a scientific language dataset of your choice

Task 2: Evaluate the quality of time series predictions with an LLM

Task 3: Explain how OmniJet- α tokenizes particles